The fourth generalized Davenport constant of C_5^3

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Abstract

We determine the remaining low-index branch in the generalized Davenport constants of the elementary abelian group C_5^3. The committed symbolic argument gives

$$5k + 10 \leq D_k(C_5^3) \leq 5k + 11 \qquad (k \geq 4)$$

and reduces the decision to whether a length-31 total-zero sequence can avoid every nonempty zero-sum subsequence of length at most five. Saturation restricts all positive multiplicities of such a sequence to $1, 2, 4$. A prior exact reduction excludes support at most 13. We give an equation-complete cover of the remaining supports 14–31: 60 multiplicity patterns split into 78 exhaustive rank/plane branches. Two exact engines with different state representations agree branch by branch across 156 runs and find no survivor. A non-generating checker reconstructs the cover, checks the bindings and rejects omitted-branch and altered-fingerprint controls. An off-host replay with a different compiler major version and native instruction target reproduces the result digest. Therefore no length-31 obstruction exists, $31 \in C_0(C_5^3)$, and

$$D_4(C_5^3) = 30.$$

The committed recurrence then yields $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$. The result is a bounded computer-assisted theorem. The replay is an independent execution of the same source, not an independent implementation, and the paper does not claim novelty of the generic zero-sum machinery or editorial authority.

1. Problem and theorem

For a finite abelian group G, let $D_k(G)$ be the least length forcing k pairwise disjoint nonempty zero-sum subsequences. The ORION-04 symbolic chain had already established the one-unit corridor

$$5k + 10 \leq D_k(C_5^3) \leq 5k + 11 \quad (k \geq 4)$$

and the conditional implication

$$D_4(C_5^3) = 30 \quad \Longrightarrow \quad D_k(C_5^3) = 5k + 10 \quad (k \geq 2).$$

It also established $D_4(C_5^3) \in \{30, 31\}$. The sole remaining numerical question was therefore the existence of a length-31 total-zero sequence with no zero sum of length one through five.

The present successor closes that question.

Theorem 1. No length-31 total-zero sequence over C_5^3 is free of nonempty zero-sum subsequences of lengths one through five. Consequently $31 \in C_0(C_5^3)$, $D_4(C_5^3) = 30$, and $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$.

The proof is computer-assisted but finite and explicitly decomposed below. Generic recurrence, localization, and zero-sum results used as premises remain donor mathematics.

2. Saturation and the finite grammar

Let S be a hypothetical obstruction. The inherited saturation theorem restricts every positive multiplicity to $1, 2, 4$. If a_1, b_2, c_4 denote the numbers of support points having those multiplicities, then

$$a_1 + b_2 + c_4 = s, \qquad a_1 + 2b_2 + 4c_4 = 31.$$

These two equations generate the outer grammar; no pattern is selected by a search outcome. The committed parent computation excludes every admissible candidate with support at most 13. The new cover contains all 42 patterns on supports 14–22 and all 18 patterns on supports 23–31.

A local projective-line enumeration supplies two further restrictions. Among the states in $\{0, 1, 2, 4\}^4$, exactly 21 are free of zero sums of lengths at most five. It follows that a multiplicity-four point is isolated on its projective line and that two multiplicity-two points cannot be collinear. Thus any two high-multiplicity support points are linearly independent.

3. Exhaustive rank and plane cover

For supports 14–22, the high-multiplicity subsequence either forces rank three through the bound $\eta(C_5^2) = 13$, supplies a rank-three basis through four multiplicity-four points, or falls into the explicit rank-two plane case when $c_4 = 3$. In the last case, two plane points normalize to e_1, e_2, the third plane direction is enumerated, and a forced point outside the plane normalizes to e_3. These cases give 51 branches.

For supports 23–31, the high-support set H is divided by rank. If $\operatorname{rank}(H) = 3$, basis extension gives one of the profiles $(2, 2, 2)$, $(4, 2, 2)$, or $(4, 4, 2)$. If its rank is two, all remaining high points lie in the normalized $\langle e_1, e_2 \rangle$ plane and an outside singleton normalizes to e_3. If $|H| = 1$, the high point extends with two singletons; if H is empty, three singletons form the basis. The $\eta(C_5^2)$ inequality removes rank-two branches whenever their high subsequence is already too long. This gives the remaining 27 branches.

The independent cover checker confirms exactly 60 patterns and 78 branches. It also rejects a cover with one branch removed. This is the completeness claim needed by Theorem 1; duplicate orbit representatives may add work but cannot remove a candidate.

4. Exact search and verification

Each engine maintains all group sums reachable at exact weights zero through five. Adding a point translates the previous exact-weight layers and rejects a partial sequence immediately if zero becomes reachable. Remaining points of a fixed multiplicity are enumerated in increasing encoded order. In plane branches, remaining doubletons are restricted to the normalized plane. Once all but one nonseed singleton are chosen, the total-zero condition uniquely forces the last singleton; distinctness, line isolation, canonical order, and the short-zero exclusion are checked before acceptance.

The primary engine represents each weight layer by a 128-bit mask and uses coordinate-mask translations. The second engine uses five 25-bit coordinate planes in AVX2 lanes with independently implemented cyclic shifts. They must agree on nodes, leaves, solution counts, and every normalized rank-two seed row—not merely on the final zero.

The frozen replay executes both engines on every branch: 78 branches times two engines, or 156 exact runs. All processes return cleanly, both representations agree on every registered fingerprint, and every solution count is zero. The non-generating verifier checks the source digests, parent binding, cover, result digest, and authority flags. Re-signed hostile controls that alter a fingerprint, omit a required branch, or promote a forbidden authority flag are rejected.

An additional replay ran the same six C sources on a different host using GCC 13.2.0 rather than GCC 14.2.0 and a different `-march=native` target. All 156 runs reproduced the exact result and cover digests. This is an independent execution and toolchain check of the frozen source. It is not an independent implementation and is not described as one.

5. Proof of the exact value

Proof of Theorem 1. Suppose a length-31 total-zero short-zero-free sequence exists. Saturation puts its multiplicities in $\{1, 2, 4\}$, so its support belongs to the equation-generated grammar. The parent theorem excludes support at most 13. The checked rank/plane decomposition partitions every remaining pattern on supports 14–31 into one of the 78 branches. Both exact engines exhaust every branch and find no solution. Hence the supposed sequence cannot exist, which is the asserted membership $31 \in C_0(C_5^3)$. Combining this with $30 \leq D_4(C_5^3) \leq 31$ gives $D_4(C_5^3) = 30$. The previously proved conditional recurrence implication now applies and gives the stated formula for every $k \geq 2$. □

6. Scope, adverse boundaries, and novelty

The earlier support-at-least-14 theorem remains a valid intermediate result but is no longer the publication ceiling. The earlier "30 or 31" terminal is historical and is superseded by Theorem 1.

The current evidence does not establish:

- a new general recurrence or localization theorem;
- novelty of standard Davenport, saturation, or projective-space machinery;
- an independent reimplementation of the exact search;
- external peer review, priority, journal fit, or acceptance;
- an algorithm-independent complexity lower bound.

The literature search found no retrieved source that states this exact value, but that search is not an exhaustive priority certificate. The manuscript therefore avoids "first" and "unique" language. A closer source may narrow the positioning without changing the checked theorem.

7. Reproducibility

The exact authority packet is `evidence/global-obstruction-v1/`. It contains the equation-generated cover, both engines, independent cover and result checkers, source and result digests, hostile controls, resource receipts, and the off-host replay receipt. The machine-verification sequence is:

```
python generate_cover.py
python independent_checker/check_static.py
python run_replay.py
python independent_checker/check_result.py
```

The source and package manifests bind the reader PDF to that evidence. A passing package checksum proves byte identity, not mathematical novelty or editorial acceptance.

8. Conclusion

The open one-unit branch for the fourth generalized Davenport constant of C_5^3 is closed. Saturation converts a hypothetical length-31 obstruction into a finite multiplicity grammar; the parent theorem excludes supports through 13; and an exact, independently checked 60-pattern and 78-branch cover excludes all remaining supports. Thus $D_4(C_5^3) = 30$, and the committed recurrence gives $D_k(C_5^3) = 5k + 10$ for every $k \geq 2$. The claim is exactly this bounded computer-assisted theorem and no broader novelty or validation claim.

Related work and novelty boundary

The generalized Davenport constants ask for the least sequence length forcing a prescribed number of pairwise disjoint nontrivial zero-sum subsequences. For rank-two groups, the exact linear formula is known, and recent inverse work studies the extremal structure at that solved boundary [Zhong 2025]. General finite abelian groups are eventually linear in the index, but the rank-three elementary setting is not covered by the rank-two formula. Nearby rank-three work also studies other zero-sum invariants rather than the same generalized Davenport constant [Zhang 2023]. Standard background is provided by the zero-sum survey of Gao and Geroldinger and the monograph of Geroldinger and Halter-Koch.

The current paper resolves the previously open numerical branch by combining the one-unit corridor and saturation/multiplicity reduction with an equation-complete support-14–31 cover and exact dual-engine exclusion. It proves `D_4(C_5^3)=30`. The residual novelty claim is confined to that bounded computer-assisted theorem and its checked decomposition; it is not a new general recurrence theory.

A primary-source review refreshed on 2026-09-01 found no retrieved source among the standard generalized-Davenport and rank-three zero-sum literature that states the exact value `D_4(C_5^3)=30`. That retrieval is not an exhaustive priority certificate. Accordingly, the paper makes no `first`, `unique`, or exhaustive-nearest-work claim; any closer source found during editorial review should narrow the positioning without changing the checked theorem statement.

References

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Positioning stop rule

The literature layer may subtract novelty but may not promote or demote the checked mathematical terminal. A priority correction changes positioning, not the exact cover or theorem.